QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

lightwalletd @ 4abfe9cd7b9b · Electric Coin Company (Zcash) — public repository (passive review)

Organization	QSI
Report date	2026-10-28 18:03 UTC
Report version	1
Audit run	14a011b1-0b2f-4c1c-b464-2a7b8ae42265 · Document audit · 2026-10-28 18:03 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Small technology company
Content SHA-256	`82414793308723f9630b1a40523260f8258d8b9c17a2b30005a7575edb8963e0`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

14	0	0	0	10
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	1
Known vulnerable component	10
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	2
Improvement recommendation	0
Insufficient data	1

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-05532	Low	Quantum-vulnerable RSA key in certificate (lightwalletd.testnet.local)	36.2
QSI-2026-05530	Low	Quantum-vulnerable RSA: role not determined (1 file(s)) preliminary	23.6
QSI-2026-05534	Low	golang.org/x/crypto v0.55.0 is affected by GO-2026-5932 (no severity rating) preliminary	20.2
QSI-2026-05535	Low	golang.org/x/crypto v0.55.0 is affected by GO-2026-6354 (CVE-2026-78662) (no severity rating) preliminary	20.2
QSI-2026-05536	Low	golang.org/x/crypto v0.55.0 is affected by GO-2026-6355 (CVE-2026-56855) (no severity rating) preliminary	20.2

Mosca's inequality (x + y > z): x = 5, y = 3, z = 12 → time available under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 0 confirmed, 1 not applicable, 0 dismissed as false positives.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/zcash/lightwalletd @ 4abfe9cd7b9b13f6093df59911639d1d9195dc09 — evidence links open the lines at this commit.

Supplied inputs

34 files, 458533 bytes in total, 1 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Small technology company" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	28	cmd/connstats.go: 39 lines scanned with 16 rule patterns and 122 registry patterns; cmd/root.go: 535 lines scanned with 16 rule patterns and 122 registry patterns; cmd/version.go: 22 lines scanned with 16 rule patterns and 122 registry patterns
tls_code	checked	2	common/generatecerts.go: TLS key-exchange settings read; frontend/rawrequest.go: TLS key-exchange settings read
certificates_and_keys	checked	2	docker/cert.key: certificates/keys parsed; docker/cert.pem: certificates/keys parsed
dependencies	checked	1	go.mod: 48 dependencies parsed (48 pinned)
osv_advisories	checked	1	48 pinned packages queried at OSV.dev; 2 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 32 (input files: C)

Architecture overview

Static review of the source tree lightwalletd at 4abfe9cd7b9b13f6093df59911639d1d9195dc09. 34 files analysed; of 121 files, 24 are test, fuzzing, benchmark or example code and were left out, 63 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. No requests were sent to the owner's systems.

Overall security posture

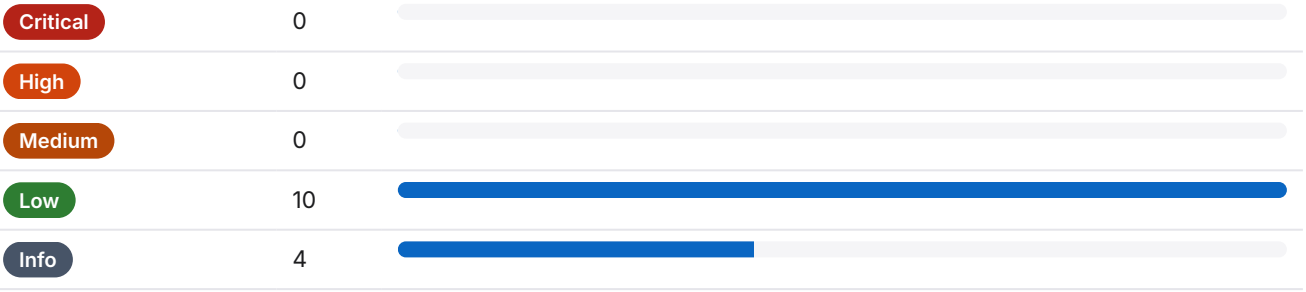
93
Indicative posture index (0–100)

37.5%
Quantum-vulnerable share of inventory

8
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-05532	Quantum-vulnerable RSA key in certificate (lightwalletd.testnet.local)	Long-term post-quantum risk	Low	High
QSI-2026-05530	Quantum-vulnerable RSA: role not determined (1 file(s))	Long-term post-quantum risk	Low	Medium, preliminary
QSI-2026-05534	golang.org/x/crypto v0.55.0 is affected by GO-2026-5932 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05535	golang.org/x/crypto v0.55.0 is affected by GO-2026-6354 (CVE-2026-78662) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05536	golang.org/x/crypto v0.55.0 is affected by GO-2026-6355 (CVE-2026-56855) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05537	golang.org/x/net v0.58.0 is affected by GO-2026-6603 (CVE-2026-78659) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05539	golang.org/x/net v0.58.0 is affected by GO-2026-6611 (CVE-2026-78669) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05540	golang.org/x/net v0.58.0 is affected by GO-2026-6612 (CVE-2026-78663) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05541	golang.org/x/net v0.58.0 is affected by GO-2026-6617 (CVE-2026-97032) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05542	golang.org/x/net v0.58.0 is affected by GO-2026-6860 (CVE-2026-78661) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05531	Private key material present in docker/cert.key	Confirmed technical vulnerability	Info	High
QSI-2026-05538	golang.org/x/net v0.58.0 is affected by GO-2026-6610 (CVE-2026-78660) (no severity rating)	Known vulnerable component	Info	Low, preliminary

QSI-2026-05543	golang.org/x/net v0.58.0 is affected by GO-2026-6866 (CVE-2026-97045) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-05533	No post-quantum capable cryptographic library detected	Insufficient data	Info	Medium, preliminary

Detailed findings

QSI-2026-05532 · CERT-QV-KEY

Low

Quantum-vulnerable RSA key in certificate (lightwalletd.testnet.local)

Long-term post-quantum risk | Confidence: High

The certificate lightwalletd.testnet.local uses a RSA 2048-bit public key, valid until 2029-12-15. RSA is broken by Shor's algorithm on a CRQC.

Affected asset	certificate:lightwalletd.testnet.local
Data source	docker/cert.pem
Detection method	X.509 public key algorithm identification.
Analyzer	certificates 1.2.0 · 2026-10-28 18:03 UTC
Priority score	36.2 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=3.13, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

For authentication, risk materializes only once a CRQC exists while the key/certificate is still trusted. Certificates are short-lived, so the main concern is the PKI and long-lived roots/ intermediates.

REPRODUCTION STEPS

```
openssl x509 -in docker/cert.pem -noout -text | grep -A2 'Public Key Algorithm'
```

EVIDENCE

certificate · docker/cert.pem · line 1 · SHA-256 4cd9b3240260015c... · File 8d44580bca050fd5...

```
Subject: CN=lightwalletd.testnet.local,O=MyOrg\, Inc.,ST=CA,C=US
Issuer: CN=lightwalletd.testnet.local,O=MyOrg\, Inc.,ST=CA,C=US
Serial: 7af3c04cf3fbe2a63724f13ab376b1c670bcaf1c
Not before: 2019-12-18T14:24:54+00:00
Not after: 2029-12-15T14:24:54+00:00
Public key: RSA 2048-bit
Signature algorithm: 1.2.840.113549.1.1.11
Signature hash: sha256
CA: True
SAN: -
SHA-256 fingerprint: 907548fce41ead924e1a866da5bb154a58b5b7e503443babd2da3e6d597341c4
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Track in the PQC inventory; prefer short certificate lifetimes; follow CA/Browser Forum and NIST PQC transition guidance for ML-DSA certificates.

RETEST PROCEDURE

Re-audit once PQC or hybrid certificates are deployed.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-05530 · CODE-QV-ASYMMETRIC

Low

Quantum-vulnerable RSA: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

2 reference(s) to RSA across 1 file(s), e.g. common/generatecerts.go:8. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:RSA:general
Data source	common/generatecerts.go
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 18:03 UTC
Priority score	23.6 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=5.0, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open common/generatecerts.go:8 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · common/generatecerts.go · line 8 · SHA-256 9b1ce2f208b801d0… · File 010d1a90f0fea3ba…

```
 7 | "crypto/rand"
 8 | "crypto/rsa"
 9 | "crypto/tls"
```

file_excerpt · common/generatecerts.go · line 21 · SHA-256 7bc59a5974adbfd5… · File 010d1a90f0fea3ba…

```
20 |
21 | privKey, err := rsa.GenerateKey(rand.Reader, 2048)
22 | if err != nil {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-05534 · DEP-ADVISORY-UNRATED

Low

golang.org/x/crypto v0.55.0 is affected by GO-2026-5932 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-5932 for Go package golang.org/x/crypto at version v0.55.0: The golang.org/x/crypto/openpgp package is unmaintained, unsafe by design, and has known security issues. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:Go/golang.org/x/crypto@v0.55.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-5932
```

EVIDENCE

tool_output · OSV.dev GO-2026-5932 · line 53 · SHA-256 ccbd80f24a2d759e...


```
{"id": "GO-2026-5932", "aliases": [], "modified": "2026-07-10T05:44:31.101996029Z", "summary": "The golang.org/x/crypto/openpgp package is unmaintained, unsafe by design, and has known security issues", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (none published yet; see the advisory for mitigations) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-5932

QSI-2026-05535 · DEP-ADVISORY-UNRATED

Low

golang.org/x/crypto v0.55.0 is affected by GO-2026-6354 (CVE-2026-78662) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6354 (CVE-2026-78662) for Go package golang.org/x/crypto at version v0.55.0: Prevent DoS on deadlocked undecided channel in golang.org/x/crypto/ssh. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/crypto/ssh.Dial: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/crypto@v0.55.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6354
```

EVIDENCE

tool_output · OSV.dev GO-2026-6354 · line 53 · SHA-256 fb187877ccc7bca3...


```
{"id": "GO-2026-6354", "aliases": ["CVE-2026-78662"], "modified": "2026-09-03T10:10:55.147673102Z",
"summary": "Prevent DoS on deadlocked undecided channel in golang.org/x/crypto/ssh", "severity":
null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.56.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6354

QSI-2026-05536 · DEP-ADVISORY-UNRATED

Low

golang.org/x/crypto v0.55.0 is affected by GO-2026-6355 (CVE-2026-56855) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6355 (CVE-2026-56855) for Go package golang.org/x/crypto at version v0.55.0: Prevent DoS on deadlocked established channel in golang.org/x/crypto/ssh. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/crypto/ssh.Dial: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/crypto@v0.55.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6355
```

EVIDENCE

tool_output · OSV.dev GO-2026-6355 · line 53 · SHA-256 4e6adace91e1ca1e...


```
{"id": "GO-2026-6355", "aliases": ["CVE-2026-56855"], "modified": "2026-09-24T10:41:42.556134596Z",
"summary": "Prevent DoS on deadlocked established channel in golang.org/x/crypto/ssh", "severity":
null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.56.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6355

QSI-2026-05537 · DEP-ADVISORY-UNRATED

Low

golang.org/x/net v0.58.0 is affected by GO-2026-6603 (CVE-2026-78659) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6603 (CVE-2026-78659) for Go package golang.org/x/net at version v0.58.0: HTTP/2 server memory exhaustion due to Trailer headers in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.connError.Error, golang.org/x/net/http2.duplicatePseudoHeaderError.Error, golang.org/x/net/http2.headerFieldNameError.Error …: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6603
```

EVIDENCE

tool_output · OSV.dev GO-2026-6603 · line 54 · SHA-256 9e43850e0da48e9d…


```
{"id": "GO-2026-6603", "aliases": ["BIT-golang-2026-78659", "CVE-2026-78659"], "modified":
"2026-10-16T09:55:47.068117441Z", "summary": "HTTP/2 server memory exhaustion due to Trailer headers
in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6603

QSI-2026-05539 · DEP-ADVISORY-UNRATED

Low

golang.org/x/net v0.58.0 is affected by GO-2026-6611 (CVE-2026-78669) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6611 (CVE-2026-78669) for Go package golang.org/x/net at version v0.58.0: Excessive CPU consumption from repeated initial window changes in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.connError.Error, golang.org/x/net/http2.duplicatePseudoHeaderError.Error, golang.org/x/net/http2.headerFieldNameError.Error …: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6611
```

EVIDENCE

tool_output · OSV.dev GO-2026-6611 · line 54 · SHA-256 c9b9f78637131224…


```
{"id": "GO-2026-6611", "aliases": ["BIT-golang-2026-78669", "CVE-2026-78669"], "modified":
"2026-10-16T09:55:38.952277866Z", "summary": "Excessive CPU consumption from repeated initial window
changes in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6611

QSI-2026-05540 · DEP-ADVISORY-UNRATED

Low

golang.org/x/net v0.58.0 is affected by GO-2026-6612 (CVE-2026-78663) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6612 (CVE-2026-78663) for Go package golang.org/x/net at version v0.58.0: Double flow control refund on HTTP/2 server streams in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.connError.Error, golang.org/x/net/http2.duplicatePseudoHeaderError.Error, golang.org/x/net/http2.headerFieldNameError.Error …: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6612
```

EVIDENCE

tool_output · OSV.dev GO-2026-6612 · line 54 · SHA-256 1471910f88c9bbf8…


```
{"id": "GO-2026-6612", "aliases": ["BIT-golang-2026-78663", "CVE-2026-78663"], "modified": "2026-10-16T09:55:54.129082194Z", "summary": "Double flow control refund on HTTP/2 server streams in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6612

QSI-2026-05541 · DEP-ADVISORY-UNRATED

Low

golang.org/x/net v0.58.0 is affected by GO-2026-6617 (CVE-2026-97032) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6617 (CVE-2026-97032) for Go package golang.org/x/net at version v0.58.0: HTTP/2 server crash due to HPACK encoder race in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.connError.Error, golang.org/x/net/http2.duplicatePseudoHeaderError.Error, golang.org/x/net/http2.headerFieldNameError.Error …: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6617
```

EVIDENCE

tool_output · OSV.dev GO-2026-6617 · line 54 · SHA-256 b948c88d959bc789…


```
{"id": "GO-2026-6617", "aliases": ["BIT-golang-2026-97032", "CVE-2026-97032"], "modified":
"2026-10-16T09:55:57.350751224Z", "summary": "HTTP/2 server crash due to HPACK encoder race in net/
http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6617

QSI-2026-05542 · DEP-ADVISORY-UNRATED

Low

golang.org/x/net v0.58.0 is affected by GO-2026-6860 (CVE-2026-78661) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6860 (CVE-2026-78661) for Go package golang.org/x/net at version v0.58.0: Avoid quadratic complexity when coalescing text nodes in html in golang.org/x/net. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code names golang.org/x/net/html.parser.parse: the affected code may be reachable.

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6860
```

EVIDENCE

tool_output · OSV.dev GO-2026-6860 · line 54 · SHA-256 1f106c45371219ba...


```
{"id": "GO-2026-6860", "aliases": ["CVE-2026-78661"], "modified": "2026-10-27T21:30:04.890809062Z",
"summary": "Avoid quadratic complexity when coalescing text nodes in html in golang.org/x/net",
"severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.62.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6860

QSI-2026-05531 · KEY-PRIVATE-IN-INPUT

Info

Private key material present in docker/cert.key

Confirmed technical vulnerability | Confidence: High

> **Analyst review (QSI review, 2026-10-23): not applicable to this code.** Self-signed development certificate generated by gen_cert.sh beside it: CN=lightwalletd.testnet.local with subjectAltName covering only localhost and 127.0.0.1. The key really is committed, so the rule is right, but it authenticates no real domain and is not a production credential. Hygiene for the maintainers rather than a vulnerability; deleting it and relying on gen_cert.sh would close it.
> automated severity: High.

A 'PRIVATE KEY' block was found at line 1 of docker/cert.key. QSI redacted the key before storage; the original file still contains it.

Affected asset	file:docker/cert.key
Data source	docker/cert.key
Detection method	PEM header detection.
Analyzer	certificates 1.2.0 · 2026-10-28 18:03 UTC
Priority score	51.5 / 100 (Small technology company)
CVSS v3.1	8.1 — CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:H/I:H/A:N

POTENTIAL IMPACT

Anyone with access to this file (repository, backups, CI logs) can impersonate the key owner or decrypt protected data.

REPRODUCTION STEPS

```
grep -n 'PRIVATE KEY-----' docker/cert.key
```

EVIDENCE

file_excerpt · docker/cert.key · line 1 · SHA-256 8ff5675e3f073e63… · File 4d417087d1f50763…

```
Line 1: -----BEGIN PRIVATE KEY----- … [REDACTED BY QSI] … -----END …
```

LIMITATIONS OF THE CHECK

QSI cannot determine whether this key is in production use. The key value was not retained.

RECOMMENDATION

Treat the key as compromised: rotate it, revoke associated certificates, remove it from version history and store keys in a secrets manager/HSM.

RETEST PROCEDURE

Search the repository and its history for private key headers; confirm the old key is revoked.

OWASP Application Security Verification Standard

QSI-2026-05538 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6610 (CVE-2026-78660) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6610 (CVE-2026-78660) for Go package golang.org/x/net at version v0.58.0: HTTP/2 transport accepts malformed framing-related headers in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/http2.Transport.NewClientConn, golang.org/x/net/http2.Transport.RoundTrip, golang.org/x/net/http2.Transport.RoundTripOpt, golang.org/x/net/http2.clientConnPool.GetClientConn, golang.org/x/net/http2.clientConnReadLoop.handleResponse, golang.org/x/net/http2.noDialClientConnPool.GetClientConn ...) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	17.1 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6610
```

EVIDENCE

tool_output · OSV.dev GO-2026-6610 · line 54 · SHA-256 1b823e53f2709379...


```
{"id": "GO-2026-6610", "aliases": ["BIT-golang-2026-78660", "CVE-2026-78660"], "modified":
"2026-10-16T09:56:04.364467528Z", "summary": "HTTP/2 transport accepts malformed framing-related
headers in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6610

QSI-2026-05543 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6866 (CVE-2026-97045) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6866 (CVE-2026-97045) for Go package golang.org/x/net at version v0.58.0: Check all ancestor namespace transitions when rendering raw text in html in golang.org/x/net. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/html.Render, golang.org/x/net/html.childTextNodesAreLiteral) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	17.1 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6866
```

EVIDENCE

tool_output · OSV.dev GO-2026-6866 · line 54 · SHA-256 102267a016bc5eef...


```
{"id": "GO-2026-6866", "aliases": ["CVE-2026-97045"], "modified": "2026-10-27T21:45:08.978077289Z",
"summary": "Check all ancestor namespace transitions when rendering raw text in html in golang.org/
x/net", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.62.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6866

QSI-2026-05533 · DEP-NO-PQC-LIBRARY

Info

No post-quantum capable cryptographic library detected

Insufficient data | Confidence: Medium | preliminary — requires further verification

Across 1 dependency manifests, QSI found crypto libraries (golang.org/x/crypto) but none from its PQC-capable list.

Affected asset	dependencies
Data source	dependency manifests
Detection method	Dependency name matching.
Analyzer	dependencies 1.2.0 · 2026-10-28 18:03 UTC
Priority score	13.4 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Readiness indicator only: PQC support may come from the platform (OpenSSL, OS, cloud KMS) rather than application dependencies.

REPRODUCTION STEPS

```
Review the dependency manifests listed in the evidence.
```

EVIDENCE

dependency · go.mod · SHA-256 08e31f77129cdbad...

```
Crypto libraries: golang.org/x/crypto
```

LIMITATIONS OF THE CHECK

Absence of evidence is not evidence of absence: system libraries and transitive dependencies may not be listed.

RECOMMENDATION

Confirm where PQC support will come from (TLS terminator, KMS/HSM, library upgrade).

RETEST PROCEDURE

Re-scan after upgrades.

FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM); FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-05532	certificate	docker/cert.pem (line 1)	4cd9b3240260015c81d32a7359585aeb0a8fc54aafe812cd312f2fa3b2bc2e7a
QSI-2026-05530	file_excerpt	common/generatecerts.go (line 8)	9b1ce2f208b801d04fec37816b6888e4ae8b469d0344091f5c8a837ee93158f9
QSI-2026-05530	file_excerpt	common/generatecerts.go (line 21)	7bc59a5974adbfd5570196de74190cfdee0b2b136760fe696a10a96bbe4f03d0
QSI-2026-05534	tool_output	OSV.dev GO-2026-5932 (line 53)	ccbd80f24a2d759ed3db1ed22b247aaeb8d2704fa45844ff83df82b05bff5e93
QSI-2026-05535	tool_output	OSV.dev GO-2026-6354 (line 53)	fb187877ccc7bca3e1e620fe5ac478176b0bfa60d40289c6b8513c063eaec76c
QSI-2026-05536	tool_output	OSV.dev GO-2026-6355 (line 53)	4e6adace91e1ca1e94bf05a7bfb237f4851e56b8c982501612e1583273e4073e
QSI-2026-05537	tool_output	OSV.dev GO-2026-6603 (line 54)	9e43850e0da48e9d00589cc2bffb372f39ec5e5a280f2f4b04520a91328fd703
QSI-2026-05539	tool_output	OSV.dev GO-2026-6611 (line 54)	c9b9f78637131224c5d4073e44c47b7cce490fc5526e3d6b0697b43bc85cab2f
QSI-2026-05540	tool_output	OSV.dev GO-2026-6612 (line 54)	1471910f88c9bbf8395a304fd82510f61567373756002de8bcdc3fbe305890cb
QSI-2026-05541	tool_output	OSV.dev GO-2026-6617 (line 54)	b948c88d959bc78948c7e3b7a746ac4da179933798e9735d02870080906cfc96
QSI-2026-05542	tool_output	OSV.dev GO-2026-6860 (line 54)	1f106c45371219bac0158f7ce44e57c439f50f0c32898616d1563f14cadb1b28
QSI-2026-05531	file_excerpt	docker/cert.key (line 1)	8ff5675e3f073e63b697c7e9001d2a28e01d7160748521613aeaa91195b89441
QSI-2026-05538	tool_output	OSV.dev GO-2026-6610 (line 54)	1b823e53f27093797b8a7909032006e1f0f818f04118ad593a4641236392faa9
QSI-2026-05543	tool_output	OSV.dev GO-2026-6866 (line 54)	102267a016bc5eef7979a90b1ddd2288ff36b09ab6fa040c808e62f3b55c1f42
QSI-2026-05533	dependency	go.mod	08e31f77129cdbad6d0a0f04c40df31cd6fe947d800a44bd559bcbe68f820e1b

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	5	5
Migration time y	3	3
Assumed years to CRQC z	12 (central)	12
x + y − z	-4 → time available under this scenario	-4 → time available under this scenario

No signature lifetime is set for the project; the data lifetime is used as an assumption.

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Small technology company”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
role not determined	RSA (1)	QSI-2026-05530	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
Grover: acceptable to NIST, below ECCG/ANSSI guidance	4
n/a	1

vulnerable (Shor)	3

Detailed findings: QSI-2026-05532, QSI-2026-05530

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — factoring Q1: quantum computation offline, classical access to the victim	breaks	RSA (3)	Breaks RSA at every key length: a longer modulus only raises the cost polynomially. Plan migration to ML-KEM (or a hybrid) for key establishment and ML-DSA / SLH-DSA for signatures; start with keys protecting data that must stay confidential for years (harvest now, decrypt later) and with long-lived signatures.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (4)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (4)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-05531 · P1 · 30 days

short-term risk reduction

Revoke/rotate the key found in docker/cert.key immediately and restrict access to the file.

Rationale	Limits the window of misuse.
Expected outcome	The exposed key is no longer trusted.
Complexity	low
Dependencies	Key owner, CA access.
Side effects	Services using the key must be redeployed.
Tests before change	Identify all systems using the key (fingerprint).
Tests after change	Confirm revocation (CRL/OCSP) and new key in use.
Closure criteria	Old key revoked, new key deployed.

QSI-2026-05531 · P1 · 60 days

root-cause fix

Move key storage to a secrets manager or HSM; add secret scanning to CI and pre-commit; purge history.

Rationale	Prevents recurrence.
Expected outcome	Keys are never committed.
Complexity	medium
Dependencies	Secrets manager.
Side effects	History rewrite affects clones.
Tests before change	Run a secret scanner over the repository.
Tests after change	Scanner reports no private keys; CI blocks new ones.
Closure criteria	Secret scanning enforced in CI; no keys in history.

QSI-2026-05534 · P3 · 60 days

root-cause fix

Triage GO-2026-5932; upgrade golang.org/x/crypto to a fixed version (none published yet; see the advisory for mitigations) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-5932, or the triage is recorded.

QSI-2026-05535 · P3 · 60 days root-cause fix

Triage GO-2026-6354 (CVE-2026-78662); upgrade golang.org/x/crypto to a fixed version (0.56.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6354 (CVE-2026-78662), or the triage is recorded.

QSI-2026-05536 · P3 · 60 days root-cause fix

Triage GO-2026-6355 (CVE-2026-56855); upgrade golang.org/x/crypto to a fixed version (0.56.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6355 (CVE-2026-56855), or the triage is recorded.

QSI-2026-05537 · P3 · 60 days root-cause fix

Triage GO-2026-6603 (CVE-2026-78659); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6603 (CVE-2026-78659), or the triage is recorded.

QSI-2026-05539 · P3 · 60 days root-cause fix

Triage GO-2026-6611 (CVE-2026-78669); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6611 (CVE-2026-78669), or the triage is recorded.

QSI-2026-05540 · P3 · 60 days root-cause fix

Triage GO-2026-6612 (CVE-2026-78663); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6612 (CVE-2026-78663), or the triage is recorded.

QSI-2026-05541 · P3 · 60 days root-cause fix

Triage GO-2026-6617 (CVE-2026-97032); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6617 (CVE-2026-97032), or the triage is recorded.

QSI-2026-05542 · P3 · 60 days root-cause fix

Triage GO-2026-6860 (CVE-2026-78661); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6860 (CVE-2026-78661), or the triage is recorded.

QSI-2026-05538 · P3 · 60 days root-cause fix

Triage GO-2026-6610 (CVE-2026-78660); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6610 (CVE-2026-78660), or the triage is recorded.

QSI-2026-05543 · P3 · 60 days root-cause fix

Triage GO-2026-6866 (CVE-2026-97045); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6866 (CVE-2026-97045), or the triage is recorded.

QSI-2026-05532 · P3 · 90 days

short-term risk reduction

Add RSA usages at certificate:lightwalletd.testnet.local to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05530 · P3 · 90 days

short-term risk reduction

Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05532 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-05530 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P1	QSI-2026-05531	Revoke/rotate the key found in docker/cert.key immediately and restrict access to the file.	short-term risk reduction

60 days

Priority	ID	Action	Type
P1	QSI-2026-05531	Move key storage to a secrets manager or HSM; add secret scanning to CI and pre-commit; purge history.	root-cause fix
P3	QSI-2026-05534	Triage GO-2026-5932; upgrade golang.org/x/crypto to a fixed version (none published yet; see the advisory for mitigations) if it applies.	root-cause fix
P3	QSI-2026-05535	Triage GO-2026-6354 (CVE-2026-78662); upgrade golang.org/x/crypto to a fixed version (0.56.0) if it applies.	root-cause fix
P3	QSI-2026-05536	Triage GO-2026-6355 (CVE-2026-56855); upgrade golang.org/x/crypto to a fixed version (0.56.0) if it applies.	root-cause fix
P3	QSI-2026-05537	Triage GO-2026-6603 (CVE-2026-78659); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-05539	Triage GO-2026-6611 (CVE-2026-78669); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-05540	Triage GO-2026-6612 (CVE-2026-78663); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-05541	Triage GO-2026-6617 (CVE-2026-97032); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-05542	Triage GO-2026-6860 (CVE-2026-78661); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.	root-cause fix
P3	QSI-2026-05538	Triage GO-2026-6610 (CVE-2026-78660); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-05543	Triage GO-2026-6866 (CVE-2026-97045); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-05532	Add RSA usages at certificate:lightwalletd.testnet.local to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-05530	Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P4	QSI-2026-05532	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-05530	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-05532	Re-audit once PQC or hybrid certificates are deployed.
QSI-2026-05530	Re-run the inventory after refactoring.
QSI-2026-05534	Re-run the dependency scan after upgrading.
QSI-2026-05535	Re-run the dependency scan after upgrading.
QSI-2026-05536	Re-run the dependency scan after upgrading.
QSI-2026-05537	Re-run the dependency scan after upgrading.
QSI-2026-05539	Re-run the dependency scan after upgrading.
QSI-2026-05540	Re-run the dependency scan after upgrading.
QSI-2026-05541	Re-run the dependency scan after upgrading.
QSI-2026-05542	Re-run the dependency scan after upgrading.
QSI-2026-05531	Search the repository and its history for private key headers; confirm the old key is revoked.
QSI-2026-05538	Re-run the dependency scan after upgrading.
QSI-2026-05543	Re-run the dependency scan after upgrading.
QSI-2026-05533	Re-scan after upgrades.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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8. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
9. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
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Appendices

A. Cryptographic inventory

Algorithm	Parameter	Location	Quantum status	Classical status
SHA-2	SHA-256	common/darkside.go:6	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	docker/cert.pem (line 1)	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	parser/block_header.go:10	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	testtools/genblocks/main.go:28	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
Crypto library golang.org/x/crypto (Go x/crypto)	v0.55.0	go.mod:53	n/a	acceptable
RSA	2048-bit	common/generatecerts.go:21	vulnerable (Shor)	acceptable
RSA		common/generatecerts.go:8	vulnerable (Shor)	acceptable
RSA	2048-bit	docker/cert.pem (line 1) — lightwalletd.testnet.local	vulnerable (Shor)	acceptable

B. JSON

qsi.report/1 · SHA-256 82414793308723f9630b1a40523260f8258d8b9c17a2b30005a7575edb8963e0

C. Input files

File	Size	SHA-256	Redactions
cmd/connstats.go	1196	`47405d57d1b057faf3be41fdcc1227795319cdcdfee56629c630e66953351311`	0
cmd/root.go	20101	`e6152b83ea0eb6273e64456e350279e7be0268da4c136648d4afeaf3e2a1cf9c`	0
cmd/version.go	517	`33979cac0db08986defae5ac62072805d733c30dd303540733b17277616d10cd`	0
common/cache.go	11643	`ba5f9da3c1492a90d48c20c15256a1bf6e354ba8bb1b58cd2fd62d9b4ddcd2a7`	0
common/common.go	25507	`c8f7603b7d32f5b076d1d61bce1be524acbe7ef20bfd35007978e638a04e9d75`	0
common/darkside.go	33254	`129a6d62e78a2bfaf78eaf34fc5fd29cadb5d074de1907669c4c63921e7dd30a`	0
common/generatecerts.go	2135	`010d1a90f0fea3babe111d4e2d463b754c4d574a16ff2840e69b6f6ef92dc19b`	0
common/logging/logging.go	1036	`5478a895e3bdb5f83bc14e71d715e55d325ceaae9030e6e00a3655460bf1f307`	0
common/mempool.go	4431	`13def4c96768f0e46aaeec5b248b801c1c267e4f6f4adc4d94ea49f7f9cc5ef2`	0
docker/cert.key	97	`4d417087d1f5076360d4260961ed5c29a58fa722450b20caf8d68765c54ab3c4`	1

docker/cert.pem	1289	8d44580bca050fd5b3a266b8e9c102a88cbdd88c5996c342b4ab8757b94d720c	0
frontend/rawrequest.go	4645	fad7ec20c623c12a517ca838059b1f6f99d76b971c42204a9c168db3cc562efb	0
frontend/rpc_client.go	3780	dd87e4a43595a01119338a5cbc010134af1eae4c90a3857ba8527deeeaec8583	0
frontend/service.go	51588	33abee53b15eece3dadbd2193eadd7c39815b8fedec4c0d589764223fcb78d47	0
go.mod	2431	2c741f2d97ee03e96e8afbeaacc5f3c7ab44cb73af859d3ad4c99930d595c108	0
hash32/hash32.go	1527	961a695822b5bc76ff5decfebc65cfab1fbbb4aada7097177ba9d3d76082c5fa	0
lightwallet-protocol/walletrpc/compact_formats.proto	6490	d8d0c8aaa5ceec7d5dcc188ef25b04011df2ec0254901620fce982fc13aeb32d	0
lightwallet-protocol/walletrpc/service.proto	16054	9352e90977cfd55ead3a0b97048c40d673ad0024d33882ae89e15727fad19138	0
main.go	89	ec16ed2f1a286086fd738c4cbb990cd6790116a39711c6a35fa405ad888a116e	0
parser/block.go	4546	3b21fb1aaf2d4d1737325894d4d2ccbff6013a29257170c67548326884a096a5	0
parser/block_header.go	8236	9fc7957a45629ec9dbadca29fee6ac6527c71a155d123c32bacf743f021a66b2	0
parser/bounds.go	3289	4303340054eaed723ee8da6e0dd6bf431d7cedb4fa9e7bf45691780ad0c7d731	0
parser/fuzz.go	354	d18aa41f9e4f215c8499d4e48add2b3075a481ab746f7f6af4144bdc19994d34	0
parser/internal/bytestring/bytestring.go	6427	cb01a5a03292979ae9d055e1ce37a137fc1c9af4f8a32b8f762040e18fe2719a	0
parser/transaction.go	24187	ff9c519ffdad866ec1ca573f742f2081df7f67c8ace42299f13513c714428265	0
testclient/main.go	4293	3f607912b23700a153a32c0f4ac95f3bb1c353a358ccf5ed2db5979f2f97eb35	0
testtools/genblocks/main.go	4934	6a7eca671d69272de2ba5471acf4d4007fbf1d6289153474360c77ba108048cc	0
testtools/zap/main.go	764	fc24949d4b8271fb936e68aeb6e6240fb258bab038a29a7eef91f615c5607de2	0
walletrpc/compact_formats.pb.go	25356	5ad9198be37bcab74f36ccbc7dca5d891632d03635f65bd70a559b46e74d04ed	0
walletrpc/darkside.pb.go	24975	51100120dc12c58486ad0eff68f564e93d70bbb2096521bef2bb5d59fa8f9b14	0
walletrpc/darkside.proto	7344	8d3582ed9efaaefeb44c2ee432191127339154b677b253350fd8ff85befff5d2	0
walletrpc/darkside_grpc.pb.go	42355	8dd2070e114aeb584628b082dce641654fb7157d4bdac5900e4ab640eb04d740	0
walletrpc/service.pb.go	62312	79749fecb58b41f9c762c374d2e4c4889665c9e450862b2a882061770b00c734	0
walletrpc/service_grpc.pb.go	51351	d5ec3b07634d3b8bbc83bb0981fc370c62b7bd24080b28d44ecc9f8f54cccd43	0